



# TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 12

## FIRST REVISION EXAMINATION, JANUARY - 2025

### MATHEMATICS

[Max. Marks : 90]

Time Allowed : 3.00 Hours]

#### PART - I

20 x 1 = 20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. If A is a  $3 \times 3$  non-singular matrix such that  $AA^T = A^T A$  and  $B = A^{-1}A^T$ , then  $BB^T =$

(1) A

(2) B

~~(3)  $I_3$~~

(4)  $B^T$

2. The rank of the matrix  $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix}$  is

~~(1) 1~~

(2) 2

(3) 4

(4) 3

3. The area of the triangle formed by the complex numbers  $z$ ,  $iz$ , and  $z + iz$  in the Argand's diagram is

~~(1)  $\frac{1}{2}|z|^2$~~

(2)  $|z|^2$

(3)  $\frac{3}{2}|z|^2$

(4)  $2|z|^2$

4. The principal argument of  $(\sin 40^\circ + i \cos 40^\circ)^5$  is

~~(1)  $-110^\circ$~~

(2)  $-70^\circ$

(3)  $70^\circ$

(4)  $110^\circ$

5. The polynomial  $x^3 - kx^2 + 9x$  has three real zeros if and only if,  $k$  satisfies

(1)  $|k| \leq 6$

(2)  $k = 0$

(3)  $|k| > 6$

~~(4)  $|k| \geq 6$~~

6. The value of  $\sin^{-1}(\cos x)$ ,  $0 \leq x \leq \pi$  is

(1)  $\pi - x$

(2)  $x - \frac{\pi}{2}$

~~(3)  $\frac{\pi}{2} - x$~~

(4)  $x - \pi$

7. If  $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$ , the value of  $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101} + y^{101} + z^{101}}$  is

~~(1) 0~~

(2) 1

(3) 2

(4) 3

8. If the function  $f(x) = \sin^{-1}(x^2 - 3)$ , then  $x$  belongs to

(1)  $[-1, 1]$

(2)  $[\sqrt{2}, 2]$

~~(3)  $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$~~

(4)  $[-2, -\sqrt{2}]$

9. The circle  $x^2 + y^2 = 4x + 8y + 5$  intersects the line  $3x - 4y = m$  at two distinct points if

(1)  $15 < m < 65$

(2)  $35 < m < 85$

(3)  $-85 < m < -35$

~~(4)  $-35 < m < 15$~~

10. Let C be the circle with centre at (1,1) and radius = 1. If T is the circle centered at (0, y) passing through the origin and touching the circle C externally, then the radius of T is equal to

(1)  $\frac{\sqrt{3}}{\sqrt{2}}$

(2)  $\frac{\sqrt{3}}{2}$

(3)  $\frac{1}{2}$

~~(4)  $\frac{1}{4}$~~



11. The eccentricity of the ellipse  $(x - 3)^2 + (y - 4)^2 = \frac{y^2}{9}$  is

(1)  $\frac{\sqrt{3}}{2}$

(2)  $\frac{1}{3}$

(3)  $\frac{1}{3\sqrt{2}}$

(4)  $\frac{1}{\sqrt{3}}$

12. If  $\vec{a}$  and  $\vec{b}$  are unit vectors such that  $[\vec{a}, \vec{b}, \vec{a} \times \vec{b}] = \frac{\pi}{4}$ , then the angle between  $\vec{a}$  and  $\vec{b}$  is

(1)  $\frac{\pi}{6}$

(2)  $\frac{\pi}{4}$

(3)  $\frac{\pi}{3}$

(4)  $\frac{\pi}{2}$

13. If  $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$ , where  $\vec{a}, \vec{b}, \vec{c}$  are any three vectors such that  $\vec{b} \cdot \vec{c} \neq 0$  and  $\vec{a} \cdot \vec{b} \neq 0$ , then  $\vec{a}$  and  $\vec{c}$  are

(1) perpendicular

(2) parallel

(3) inclined at an angle  $\frac{\pi}{3}$

(4) inclined at an angle  $\frac{\pi}{6}$

14. If the direction cosines of a line are  $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$  then

(1)  $c = \pm 3$

(2)  $c = \pm \sqrt{3}$

(3)  $c > 0$

(4)  $0 < c < 1$

15. The number of positive zeros of the polynomial  $\sum_{j=0}^n {}^nC_j (-1)^j x^j$  is

(1) 0

(2) n

(3)  $< n$

(4) r

16. If  $z + \frac{1}{z} + 1 = 0$ , then  $z^{2003} + \frac{1}{z^{2003}}$  is equal to-

(1) 1

(2) -1

(3) 0

(4) None of these

17. If the system of equations  $x + 2y - 3z = 1, (k+3)z = 3, (2k+1)x + z = 0$  is inconsistent, then the value of k is

(1) -3

(2)  $\frac{1}{2}$

(3) 0

(4) 2

18. Latus rectum of ellipse  $4x^2 + 9y^2 - 8x - 36y + 4 = 0$  is

(1)  $8/3$

(2)  $4/3$

(3)  $\frac{\sqrt{5}}{3}$

(4)  $16/3$

19. If  $\sin^{-1} \frac{1}{3} + \sin^{-1} \frac{2}{3} = \sin^{-1} x$ , then x is equal to

(1) 0

(2)  $\frac{\sqrt{5}-4\sqrt{2}}{9}$

(3)  $\frac{\sqrt{5}+4\sqrt{2}}{9}$

(4)  $\frac{\pi}{2}$

20. The value of  $z = (1+i)^8 + (1-i)^8$  is.

(1) 16

(2) -16

(3) 32

(4) -32

## PART - II

1. Answer any 7 questions

7 x 2 = 14

2. Each question carries 2 marks

3. Question number 30 is compulsory

21. Verify the property  $(A^T)^{-1} = (A^{-1})^T$  with  $A = \begin{bmatrix} 2 & 9 \\ 1 & 7 \end{bmatrix}$ .

22. Construct a cubic equation with roots  $2, \frac{1}{2}$  and 1.

23. Examine the position of the point (2,3) with respect to the circle  $x^2 + y^2 - 6x - 8y + 12 = 0$ .



24. Let  $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ ,  $\vec{b} = \hat{i}$  and  $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ . If  $c_1 = 1$  and  $c_2 = 2$ , find  $c_3$  such that  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  are coplanar.
25. A chemist has one solution which is 50% acid and another solution which is 25% acid. How much each should be mixed to make 10 litres of a 40% acid solution? (Use Cramer's rule to solve the problem).
26. Simplify:  $i^{59} + \frac{1}{i^{59}}$
27. If  $k$  is real, discuss the nature of the roots of the polynomial equation  $2x^2 + kx + k = 0$  in terms of  $k$ .
28. Identify the type of conic section for  $3x^2 + 3y^2 - 4x + 3y + 10 = 0$
29. Find the magnitude and the direction cosines of the torque about the point  $(2, 0, -1)$  of a force whose line  $2\hat{i} + \hat{j} - \hat{k}$  action passes through the origin.
30. If  $\vec{x} \cdot \vec{a} = 0$ ,  $\vec{x} \cdot \vec{b} = 0$ ,  $\vec{x} \cdot \vec{c} = 0$ ,  $\vec{x} \neq 0$ . Prove that  $\vec{a}$ ,  $\vec{b}$ ,  $\vec{c}$  are Coplanar.

## PART - III

1. Answer any 7 questions
2. Each question carries 3 marks
3. Question number 40 is compulsory

7x3=21

31. Find the rank of the matrix  $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$
32. Find the square root of  $6-8i$ .
33. Find the domain of  $g(x) = 2 \sin^{-1}(2x-1) - \frac{\pi}{4}$ .
34. Find the equation of the hyperbola with vertices  $(0, \pm 4)$  and foci  $(0, \pm 6)$ .
35. Find the equation of the plane passing through the line of intersection of the planes  $x + 2y + 3z = 2$  and  $x - y + z + 11 = 3$ , and at a distance  $\frac{2}{\sqrt{3}}$  from the point  $(3, 1, -1)$ .
36. Find the condition on  $a$ ,  $b$ , and  $c$  so that the following system of linear equations has one parameter family of solution  $x + y + z = a$ ,  $x + 2y + 3z = b$ ,  $3x + 5y + 7z = c$ .
37. If  $z = x + iy$ , find the rectangular form of  $\text{Im}(3z + 4\bar{z} - 4i)$
38. Find the value of  $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right) + \cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)$
39. If the straight lines  $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{m^2}$  and  $\frac{x-3}{1} = \frac{y-2}{m^2} = \frac{z-1}{2}$  are coplanar, find the distinct real values of  $m$ .
40. Find the value of  $4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99}$

## PART - IV

1. Answer all the questions
2. Each question carries 5 marks

7x5=35

41. a) A boy is walking along the path  $y = ax^2 + bx + c$  through the points  $(-6, 8)$ ,  $(-2, -12)$ , and  $(3, 8)$ . He wants to meet his friend at  $P(7, 60)$ . Will he meet his friend? (Use Gaussian elimination method.)
- (OR)



b) If  $A = \begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$ , find the products AB and BA and hence solve the system of equations

$$x - y + z = 4, x - 2y - 2z = 9, 2x + y + 3z = 1.$$

42. a) If  $\frac{1+z}{1-z} = \cos 2\theta + i \sin 2\theta$ , show that  $z = i \tan \theta$ .

(OR)

b) Let  $z_1, z_2$ , and  $z_3$  be complex numbers such that  $|z_1| = |z_2| = |z_3| = r > 0$  and  $z_1 + z_2 + z_3 \neq 0$ .

Prove that  $\left| \frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{z_1 + z_2 + z_3} \right| = r$

43. a) Solve the following equations :  $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$ .

(OR)

b) Solve :  $(2x - 1)(x + 3)(x - 2)(2x + 3) + 20 = 0$

44. a) Find the equation of the circle passing through the points (1,1), (2, -1) and (3,2).

(OR)

b) A tunnel through a mountain for a four lane highway is to have a elliptical opening. The total width of the highway (not the opening) is to be 16m, and the height at the edge of the road must be sufficient for a truck 4m high to clear if the highest point of the opening is to be 5m approximately. How wide must the opening be?

45. a) Find the value of  $\sin \left( \tan^{-1} \left( \frac{1}{2} \right) - \cos^{-1} \left( \frac{4}{5} \right) \right)$

(OR)

b) Prove that  $\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}$ ,  $-1 < x < 1$

46. a) Prove by vector method that  $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$ ,

(OR)

b) Find the parametric form of vector equation of the straight line passing through (-1,2,1) and parallel to the straight

line  $\vec{r} = (2\hat{i} + 3\hat{j} - \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k})$  and hence find the shortest distance between the lines.

47. a) Prove that the line  $5x + 12y = 9$  is a tangent to hyperbola  $x^2 - 9y^2 = 9$ . Also find the point of contact.

(OR)

b) If  $a = \cos 2\alpha + i \sin 2\alpha$ ,  $b = \cos 2\beta + i \sin 2\beta$ ,  $c = \cos 2\gamma + i \sin 2\gamma$ , prove  $\sqrt{abc} + \frac{1}{\sqrt{abc}} = 2 \cos(\alpha + \beta + \gamma)$  and

$$\frac{a^2 b^2 + c^2}{abc} = 2 \cos 2(\alpha + \beta - \gamma).$$